PHYSICAL CHALLENGE

Overclocking challenges sheer computing prowess. Get in tune with the full potential of your PC, but remember not to let the PC lose its cool

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Airflow

Getting cool air to warm components is the most fundamental cooling principle to adhere to. A traditional ATX cabinet has two fans, one to suck in cool air and the other to expel hot air. But the purpose can be defeated by the presence of loose connectors and cables inside. To maximise the flow of air within the cabinet, carefully twist, tie or tape up all loose cables and place them away from components such as the CPU, hard disk and RAM.

Heatsinks

Knowing that the CPU is the most overclocked part of the computer, getting a high quality fan/heatsink for the chip is essential. Take a serious look at some powerful fan/heatsink combos made by the likes of Thermaltake, Evercool, Akasa and others. A high quality fan would cost you at least Rs 700, but considering its importance, don't even think of settling for anything less.

Spread it!

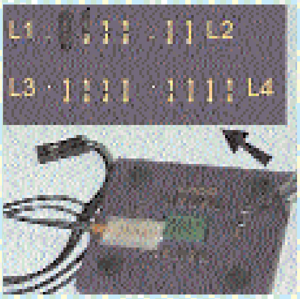
It's not sandwich spread, but a thin layer of this stuff goes a long way in effectively dispersing heat away from critical components. Thermal paste is the substance that is applied between the heatsink and for example, a CPU. You can even use it with memory modules and video cards. Just make sure you use a very thin layer, otherwise, you'll completely defeat the purpose of using it.

Pencil Surgery

For AMD processor owners, there is a simple trick you can perform to confer your chip with the potential to run at speeds a lot higher than you ever thought possible. Of course, you can overclock by increasing the FSB, but the fun would pale in comparison to what you could enjoy by going all the way. All you need to 'unlock' an Athlon or a Duron CPU is something very simple—a pencil, preferably a 0.5 mm clutch pencil. Do it as we spell it here:

■ Remove your CPU and place it on a flat surface as shown in the picture. Observe the CPU closely and look out for four electrical contacts, or bridges, labeled L1. If the contacts are broken, pick up the pencil and fill in the gaps. If the contacts are in place, skip the following steps and directly attempt to overclock the chip.

■ With a steady hand, work your way across the bridges and connect them by rubbing the pencil back and forth about a dozen times or until the bridges are dark black and not the usual gold colour. Do the same for all L1 bridges and make sure that the pencil marks do not overlap each other, otherwise the trick won't work.



Ensure you don't draw lines that overlap.

■ Now that you have this licked, you're ready to overclock the chip to its limits. Instead of using the feeble FSB method, you can now increase clock speed by stepping up the multiplier.

A typical Duron 600 runs at a 100 MHz FSB with a multiplier of 6, which gives you its 600 MHz clock speed. Now if you were to increase the multiplier to 9, do the math and you'll come up with a 900 MHz overclock. That's a 50 per cent increase in clock speed.

Remember, in order to carry out this procedure, your motherboard must support multiplier adjustments. Some models let you change it through DIP switches while others have it in the system BIOS. You must refer to your motherboard's manual on how to change the multiplier value.

Intel chips

If you're the owner of a PC with 'Intel Inside', the only avenue you have in order to overclock the chip is driving up FSB speeds. Unfortunately, each and every chip made by Intel comes multiplier locked with no avenue to circumvent it. But unlike AMD, processors from Intel